BELGIF
BELGIAN INTEROPERABILITY FRAMEWORK

ICEG Person

Thematic Workshop 1

Welcome!

Monday 21 September 2026
Marie-Elisabeth Belpaire (03.N.08)

From 10:00am till 12:30pm



Practical information

Iterative and consensus-based approach, following the U−1 decision rule

Quick polls in the room:

- +1 = agree
- −1 = disagree
- 0 = no opinion



Raise hand or use chat to ask a question

Workshop reports will be made publicly available on GitHub

Recording ?

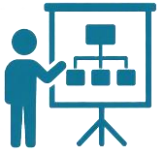


Goal of this session

ICEG – Person | Thematic workshop 1



Introduction of the ICEG Person trajectory



Presentation of the draft model



Feedback on the draft model and capturing the input

Planning

Time	Item
10:00 - 10:05	Welcome and practical information
10:05 - 10:15	Planning
10:15 - 10:20	Rationale for the trajectory and context
10:20 - 10:30	Approach and use cases
10:30 - 11:00	Registration of a person
11:00 - 11:15	Break
11:15 - 11:30	Registration of a Noble Title
11:30 - 12:15	Change of name and registered sex
12:15 - 12:30	Q&A and next steps

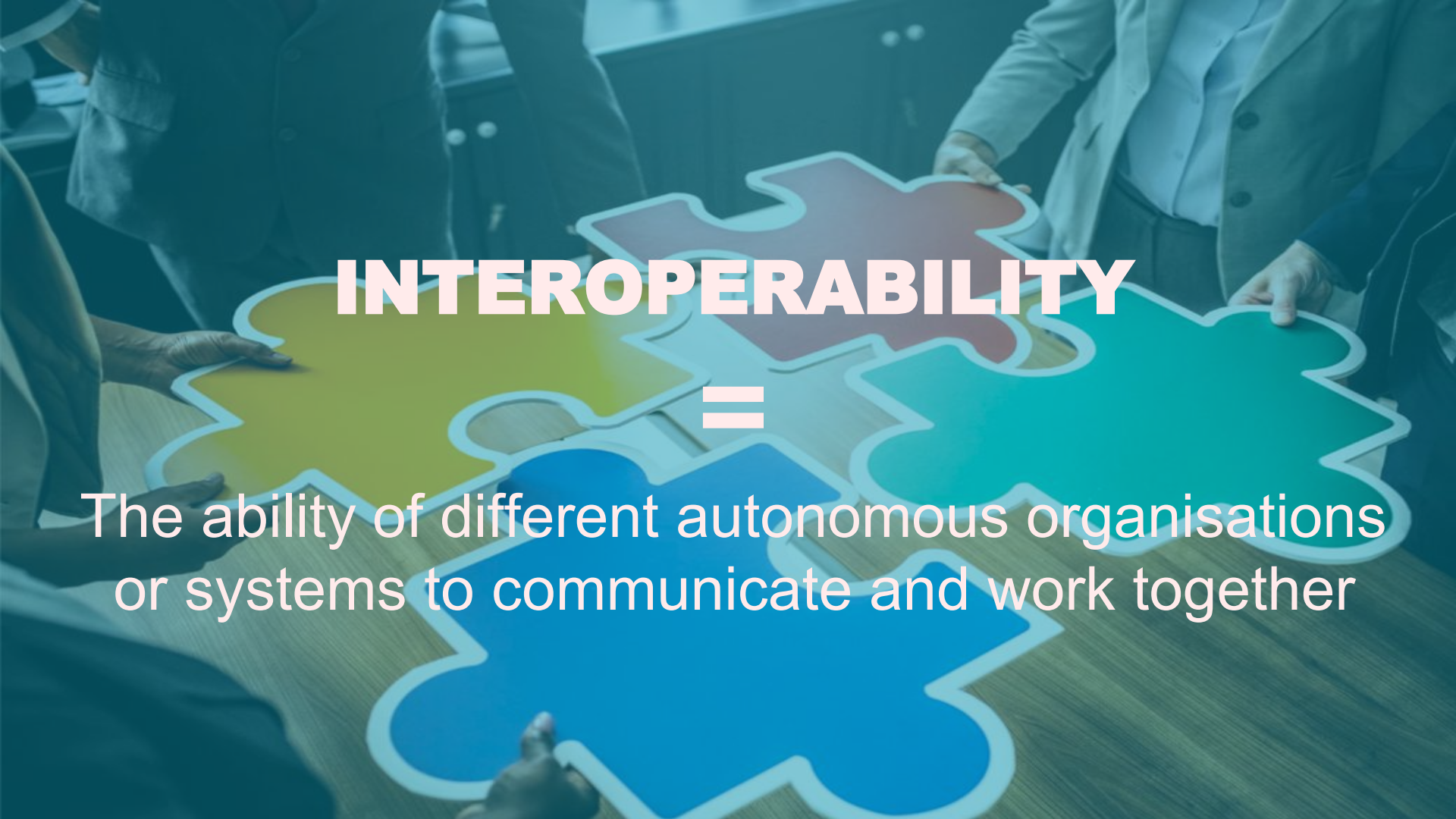
Tour de table



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Rationale for the trajectory and context

10:15 - 10:20


INTEROPERABILITY
=
The ability of different autonomous organisations or systems to communicate and work together

Context

- Personal data is currently fragmented across administrations: different structures, definitions, identifiers
- This fragmentation complicates data exchange between entities
- Different ways of representing and interpreting the same data add to this problem: i.e.: combining KSZ and RRN data is often difficult.



Goal

Establish ONE shared, consensus-based, semantic ICEG model for person data in Belgium.

👉 **Outputs:**

Artefact	Description
Vocabulary	Formal semantic vocabulary defining classes and properties for person data
Documentation	Human-readable documentation of terms, definitions, and usage guidelines
UML Diagram	Visual representation of the information model and relationships
SHACL Validation Rules	Validation constraints for data quality assurance
JSON-LD Context File	Context file enabling JSON-based Linked Data serialization
OpenAPI/Swagger 3.0 Spec	API specification for integrators/consumers to build or adapt implementation models
Code Lists	Standardized reference data (nationalities, civil status codes, register types, etc.)

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Approach and scope

10:20 - 10:30

Approach



Iterative design of the model based on use cases



We align as much as possible with existing standards (W3C, ISO, SEMIC, OSLO, etc.)



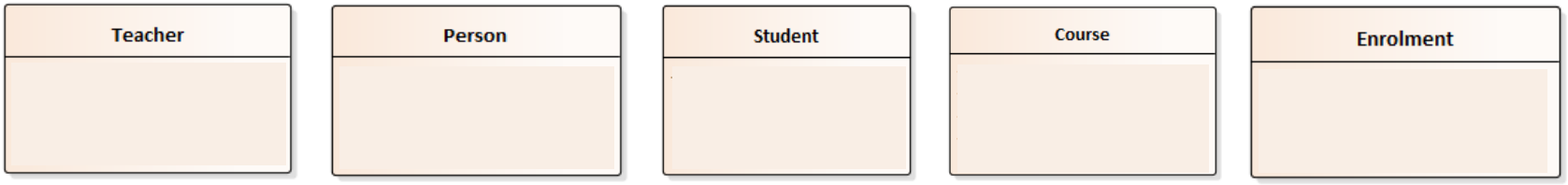
We define things ourselves where necessary

UML - Unified Modeling Language

USE CASE: A teacher gives a course, and students enroll for that course.

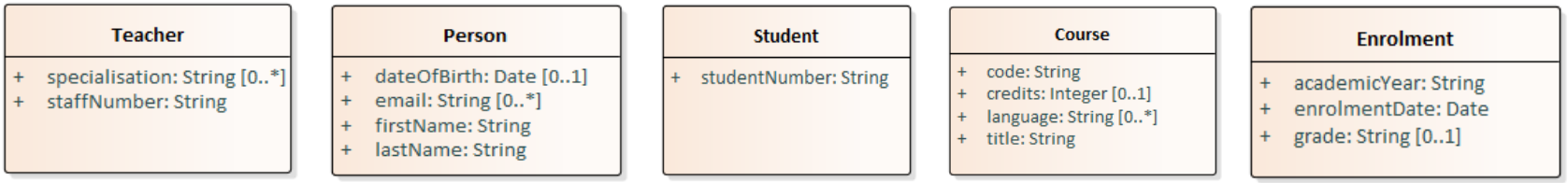
1. Concepts (classes)

Concept	Meaning
Person	A human being (general concept)
Teacher	A person who teaches
Student	A person who follows a course
Course	A subject that is taught (e.g. Mathematics)
Enrolment	The registration of a student for a course



2. Attributes

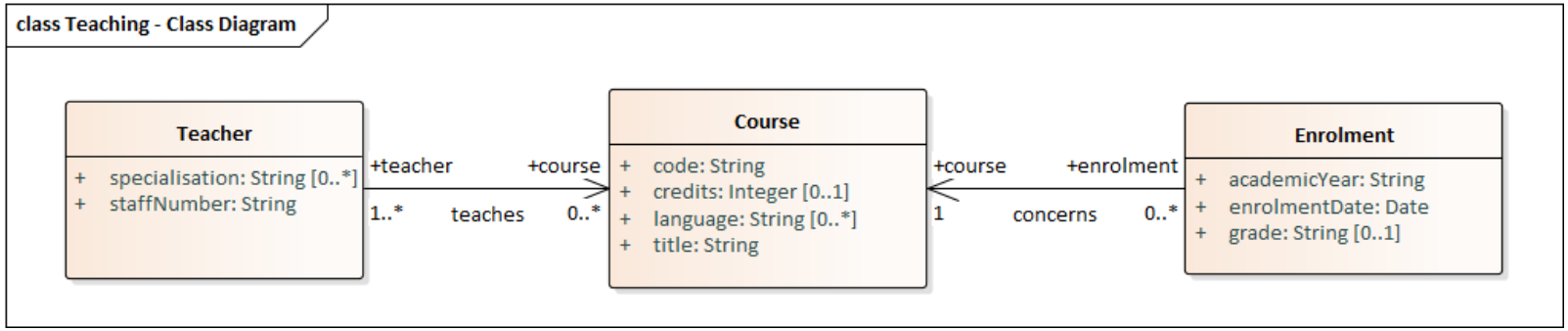
Class	Attributes
Person	first name, last name, date of birth, e-mail
Teacher	staff number, specialisation
Student	student number
Course	code, title, number of credits, language
Enrolment	enrolment date, academic year, grade



3. Relations (associations)

Relations	Meaning
Teacher - Course	a teacher teaches 0..* courses; a course is taught by 1..* teachers
Student - Enrolment	a student has 0..* enrolments
Enrolment - Course	an enrolment concerns exactly 1 course

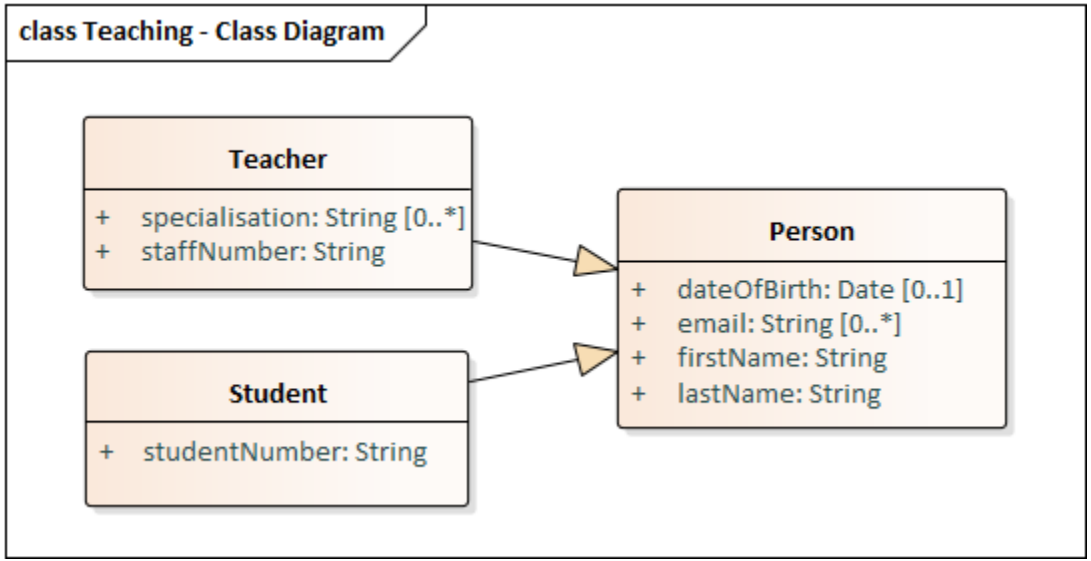
1	exactly one (mandatory)
0..1	zero or one (optional)
0..* or *	zero or more
1..*	one or more (at least one)



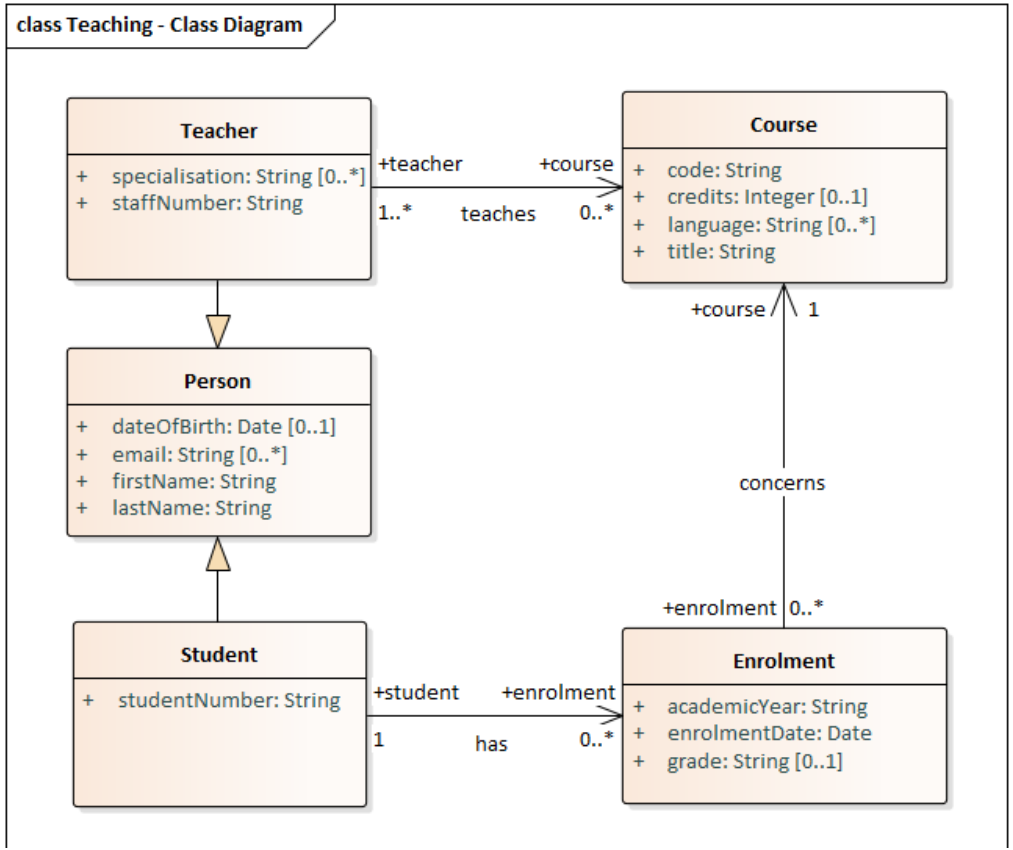
4. Generalisation

Teacher and Student are subclasses of Person. They inherit first name, last name, date of birth and e-mail, and add their own attributes.

👉 A person can even be both, a teacher who follows a course.



5. Model


class Teaching - Class Diagram
Teacher
+ specialisation: String [0..*]
+ staffNumber: String
+teacher
+course
1..*
teaches
0..*
Course
+ code: String
+ credits: Integer [0..1]
+ language: String [0..*]
+ title: String
+course
1
Person
+ dateOfBirth: Date [0..1]
+ email: String [0..*]
+ firstName: String
+ lastName: String
concerns
+enrolment
0..*
Student
+ studentNumber: String
+student
+enrolment
1
has
0..*
Enrolment
+ academicYear: String
+ enrolmentDate: Date
+ grade: String [0..1]

Scope ICEG-Person trajectory

✅ In-Scope	❌ Out-of-Scope
Natural Person: identity, identifiers (SSIN), name, date/place of birth, gender, nationality, domicile, residence, contact information, civil status, legal capacity	Domain-specific person profiles (health, taxation, profession, etc.)
Registered Person: registration in official registers (National Register, BIS register)	Authentication and authorisation mechanisms lines
Links to Other Domains: Address (aligned with BeSt-Address), possible relation to organization (via CBE), jurisdiction, household composition, filiation	Internal database properties, schemas, or storage implementations

Scope TW1

- Identification Number (TI 000)
- Reference File (TI 002)
- Name, First Names, Titles, Aliases (TI 010, 011, 012, 013)
- Birth Information (TI 100, 101)
- Gender (TI 004)

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Use Case 1
Registration of a person

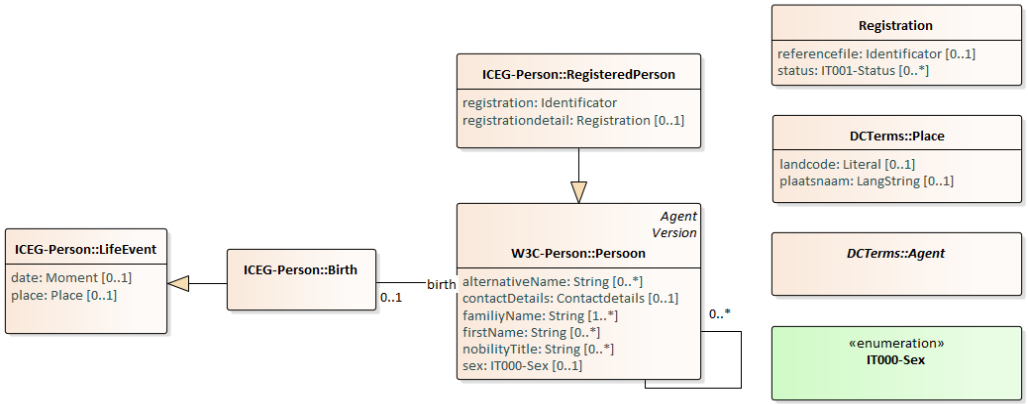
10:30 - 11:00

Registration of a person

*Julie is **born** in the **hospital of Leuven**. Her **parents** declare the birth to the local authority. The **municipality** registers her **name**, her **first names** and her **birth data**, and an **identification number** is assigned to **her**. All of this is documented in a **reference dossier**.*

Declaration of Birth & Identification

Julie is born. Her parents declare the birth to the local authority. The municipality prepares to register her Name and Birth information, which will trigger identification. Based on her registered Name and Birth data, the National Register automatically assigns Julie a unique Identification Number.



What is a Declaration?

The formal act by which parents notify the local authority of a birth, triggering the registration of Name and Birth information.

What is Name (TI 010)?

The information type containing a person's family name and first names, encoded using codes assigned by the National Register.

What is Birthplace (IT 100)?

The information type recording the date, act number, hour and place of birth. For births in Belgium the NIS-code of the municipality; for births abroad, the place name + country code.

What is Birth Date (IT 101)?

The information type holding the date of birth and the type of supporting document used to prove it. This date must always match the date encoded in the person's identification number.

What is an IdentificationNumber?

The unique 11-digit number (SSIN/NI) automatically assigned upon a person's first registration, computed from date of birth, gender, and a control number. The identification number can never be reused once cancelled.

BREAK



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Use Case 2
Registration of a Noble Title and Pseudonym

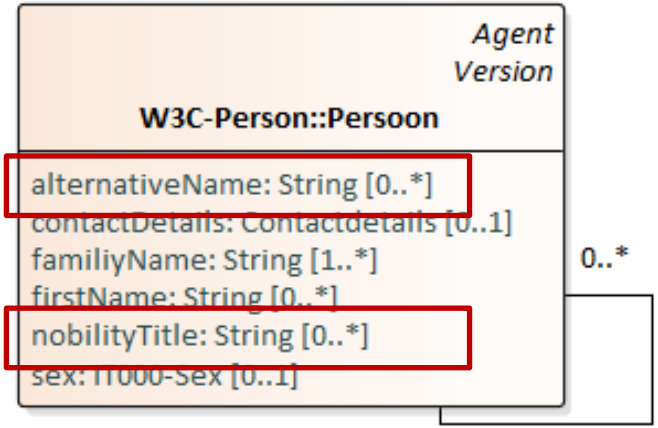
11:15 - 11:30

Registration of a Noble Title



Registration Noble Title & Pseudonym

Philippe is granted the noble title of "Knight" by Royal Decree. The municipality registers his new title in the National Register, referencing the decree that justifies the change. Later in his artistic career, Philippe also adopts a pseudonym, which is registered separately and remains independent of his legal name and title.



What is a Noble Title (IT 012)?

An honorary distinction assigned to a person, i.e.: Knight, Baron, etc.) and the date on which the title was obtained. Each new registration is a separate IT012 entry, it does not overwrite the previous one, so earlier titles remain present in the dossier.

What is a Pseudonym (TI 011)?

An assumed name, independent of the legal name, most often adopted by artists. It has no legal value for official identification but can still be protected like a name.

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Use Case 3
Change of name and registered sex

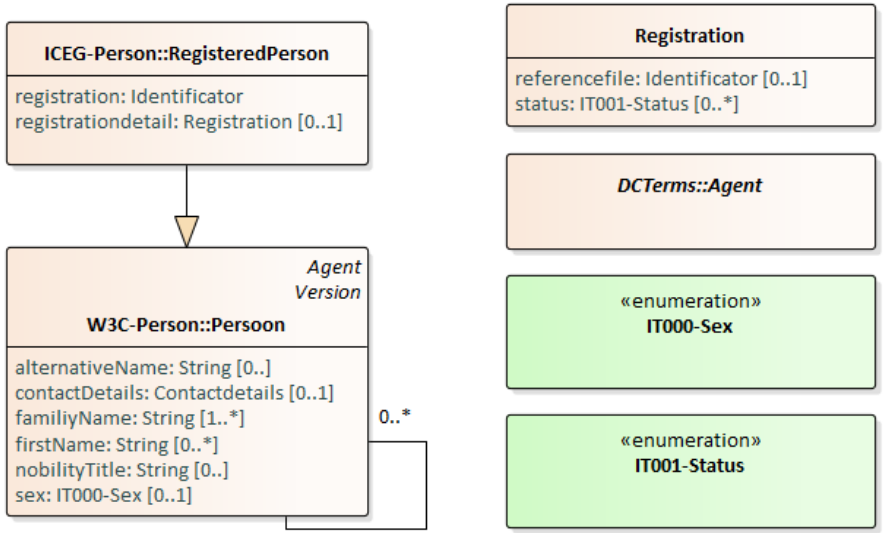
11:30 - 12:15

Change of name and registered sex

*Jonas has been registered at birth under a different first name and a different sex. After a procedure with the competent authority, his **first name is changed and his registered sex is adapted**. The previous situation must remain **traceable.***

Change of FirstName & Registered Sex

Jonas was registered at birth under a different first name and a different sex. After a procedure with the competent authority, his first name is changed and his registered sex is adapted. The previous situation (former name and sex) must remain traceable in the National Register via a reference file.



What is NameModification (TI 013)?

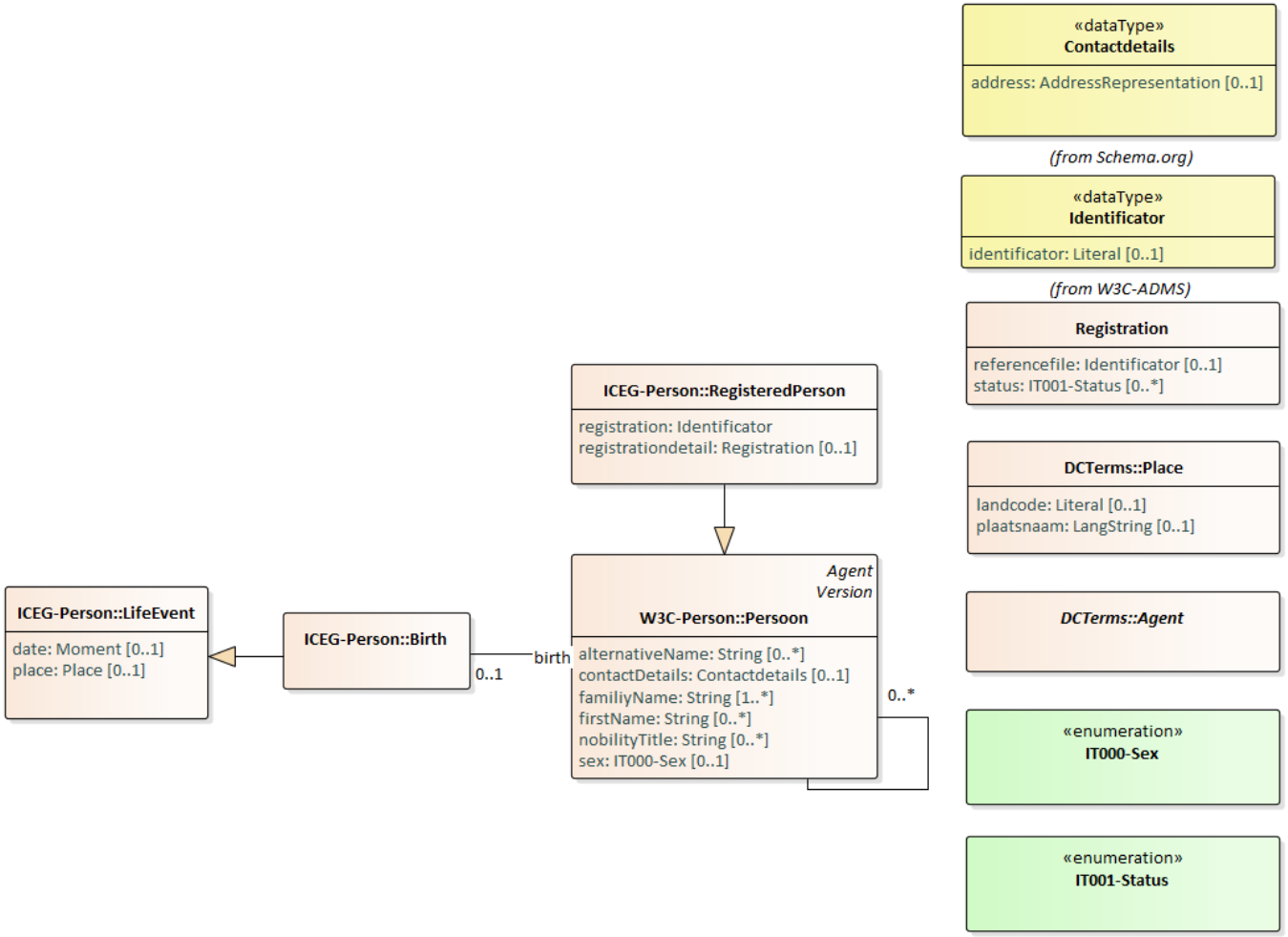
The information type recording the reason for a change or correction of name/first names. It's always dated and justified (reasonCode enumeration). TI 013 explicitly covers modifications to name, first names, AND title of nobility with the same structure (date + reason code + comment). A TI 013 entry always requires a same-dated TI 010 (or TI 012) entry already present in the file. The reason can never exist without the corresponding value change.

What is GenderModification (TI 004)?

The information type recording the modification of a person's registered sex, following a declaration to the civil status officer under Art. 135/1 of the Civil Code.

What is a ReferenceFile (TI 002)

The information type recorded in a cancelled dossier, linking it to the new file that replaces it. It contains the date of annulment, the identification number of the new dossier, and a justification code explaining the reason for cancellation. The new dossier does not automatically inherit the content of the previous one; the ReferenceFile only ensures traceability between the old and new file.


«dataType»
Contactdetails
address: AddressRepresentation [0..1]
(from Schema.org)
«dataType»
Identificator
identificator: Literal [0..1]
(from W3C-ADMS)
Registration
referencefile: Identificator [0..1]
status: IT001-Status [0..*]
ICEG-Person::RegisteredPerson
registration: Identificator
registrationdetail: Registration [0..1]
DCTerms::Place
landcode: Literal [0..1]
plaatsnaam: LangString [0..1]
Agent
Version
W3C-Person::Persoon
alternativeName: String [0..*]
contactDetails: Contactdetails [0..1]
familiyName: String [1..*]
firstName: String [0..*]
nobilityTitle: String [0..*]
sex: IT000-Sex [0..1]
DCTerms::Agent
ICEG-Person::LifeEvent
date: Moment [0..1]
place: Place [0..1]
ICEG-Person::Birth
birth
0..1
0..*
«enumeration»
IT000-Sex
«enumeration»
IT001-Status

Q&A



Next Steps



Processing all input from this thematic working group



Circulating a report of this working group –> any feedback is welcome



Incorporating the input of this session into the current draft model

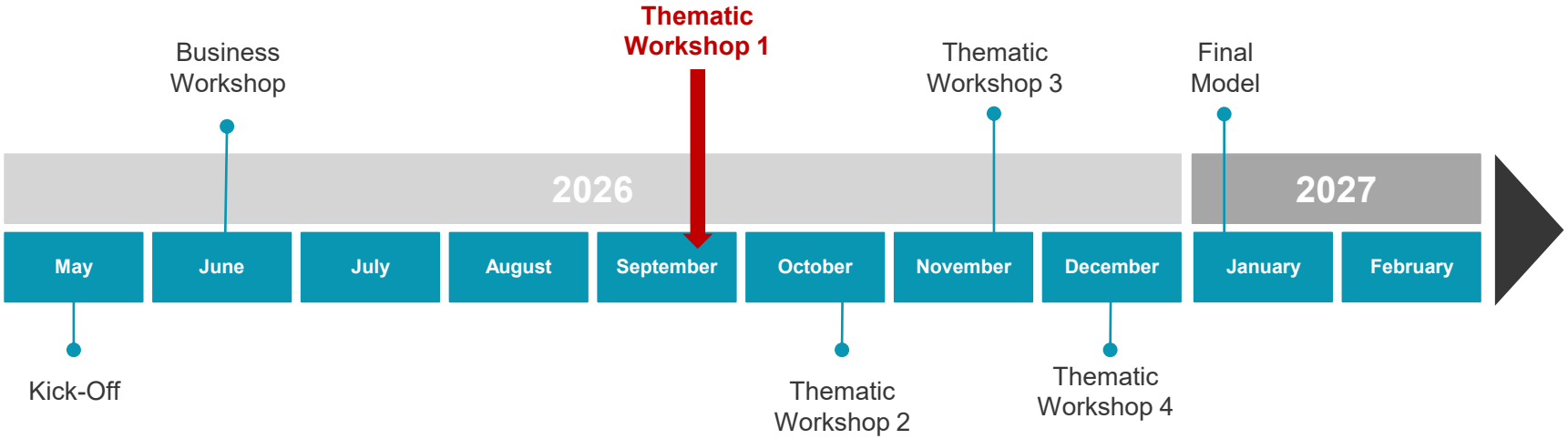


Capturing further feedback via GitHub



Publishing a first version of the semantic model

Timeline



Thematic workshop Workshop 2: Address and Residence
Date: Monday 19 October 2026, 10:00 – 12:30
👉 Register via the following link: Topics & registration workshop 2.

Feedback & Collaboration

Feedback can be given by e-mail to the following people:

- valy.vandenbroeck@vlaanderen.be
- dylan.vanassche@vlaanderen.be
- jitse.decock@vlaanderen.be

Feedback/input can be given via GitHub by creating ***Issues***:

https://github.com/belgif/thematic/issues

👉 Public review period: ?

The minutes of this session will be shared on GitHub within three working days following the Thematic Workshop. You are invited to submit your comments within five working days of publication.

THANK YOU !